

KEMIJARVI, Finland

WMO: 028140

Lat: 66.716N

Lon: 27.159E

Elev: 682

StdP: 14.34

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-18.9	-30.1	1.0	-24.4	-23.8	1.5	-18.7	22.6	23.6	20.4	24.6	2.5	320	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	77.5	61.0	73.5	59.9	69.9	58.1	64.5	72.7	62.3	70.2	60.2	67.0	7.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.6	84.0	67.0	59.3	77.2	64.6	57.3	71.8	62.8	29.9	72.6	28.3	70.1	26.8	66.8	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.4	14.4	-30.9	80.7	4.3	4.9	-34.0	84.3	-36.5	87.1	-38.8	89.9	-41.9	93.5		
(5)			-30.5	67.5	4.5	2.8	-33.8	69.5	-36.4	71.1	-38.9	72.7	-42.2	74.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.2	8.7	12.0	19.0	30.0	43.2	52.0	58.6	53.9	46.0	32.6	23.7	16.8		
	DBStd	19.16	13.26	14.30	10.36	7.81	8.52	7.45	6.32	6.31	5.99	7.58	9.61	12.66		
	HDD50	6761	1281	1065	962	599	248	63	8	32	146	540	789	1028		
	HDD65	11650	1746	1485	1427	1049	677	396	213	349	571	1005	1239	1493		
	CDD50	610	0	0	0	0	36	122	275	152	25	0	0	0		
	CDD65	26	0	0	0	0	0	6	16	4	0	0	0	0		
	CDH74	297	0	0	0	0	15	59	183	40	0	0	0	0		
	CDH80	45	0	0	0	0	0	9	29	7	0	0	0	0		
	WSAvg	6.6	5.9	6.4	7.5	6.5	7.2	7.2	6.0	6.6	6.5	6.2	6.2	6.6		
	PrecAvg	22.9	1.7	1.4	1.3	1.1	1.9	2.4	3.0	2.4	2.2	2.0	1.8	1.8		
	PrecMax	31.2	2.9	2.8	2.1	2.5	3.8	3.9	4.4	5.4	4.4	3.6	3.1	3.0		
(15) Precipitation	PrecMin	17.7	0.7	0.2	0.4	0.4	0.6	0.6	0.8	1.1	0.3	0.8	0.6	0.6		
	PrecStd	3.8	0.7	0.7	0.5	0.6	0.7	1.0	1.1	1.1	1.1	0.8	0.7	0.7		
	DB	33.1	37.3	41.3	56.2	76.1	80.6	84.2	79.8	64.7	50.1	41.2	36.9			
	MCWB	32.2	35.4	35.9	44.4	55.2	61.5	64.6	66.2	56.3	48.7	40.5	35.9			
	2% DB	31.2	33.8	37.2	47.6	70.7	74.2	79.0	72.6	61.4	46.7	38.5	34.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	30.8	32.9	33.3	39.7	54.8	60.0	62.4	60.7	54.3	44.5	37.7	33.7			
	5% DB	29.2	32.2	34.6	43.8	63.7	69.5	74.9	68.8	58.6	44.7	36.2	32.9			
	MCWB	28.8	31.5	31.8	37.4	51.7	56.0	61.3	59.0	52.9	43.2	35.7	32.4			
	10% DB	26.6	30.0	32.6	40.9	58.1	65.2	71.5	65.0	55.9	42.6	34.3	31.6			
	MCWB	26.2	29.3	30.8	35.5	48.3	54.5	60.0	57.2	52.0	41.3	33.8	31.3			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	32.4	35.7	36.6	45.3	57.5	64.6	68.2	66.7	58.7	49.5	41.0	36.0			
	MCDB	32.8	37.2	40.0	54.7	70.6	73.8	75.7	76.5	61.9	50.1	41.1	36.6			
	2% WB	30.9	33.1	34.1	41.1	55.3	62.3	65.4	62.9	55.6	45.0	37.8	33.8			
	MCDB	31.2	33.7	36.2	46.7	68.5	72.9	72.6	70.2	59.3	46.0	38.1	34.1			
	5% WB	28.9	31.7	32.5	38.1	52.7	59.0	63.3	60.4	54.0	43.2	35.8	32.5			
(32) Mean Daily Temperature Range	MCDB	29.2	32.1	34.0	42.3	63.0	65.9	71.0	65.1	57.2	44.5	36.3	32.9			
	10% WB	26.2	29.4	31.2	36.3	49.2	56.2	61.3	58.6	52.5	41.5	34.0	31.2			
	MCDB	26.6	30.0	32.6	39.7	56.6	62.5	69.0	63.2	55.4	42.5	34.3	31.6			
	5% DB	13.9	13.7	17.4	15.9	16.3	15.1	16.0	15.3	12.6	8.7	9.5	11.7			
	MCDBR	12.4	9.9	13.8	19.3	26.7	22.3	23.3	21.6	19.1	10.3	7.4	8.5			
(36) Clear-Sky Solar Irradiance	MCWBR	12.5	9.4	11.2	12.1	13.3	10.2	9.8	10.9	12.6	8.7	7.3	8.3			
	5% WB	12.9	9.8	10.0	16.8	24.3	18.6	19.5	16.5	14.4	9.4	7.3	8.4			
	MCDBR	13.0	9.4	8.6	11.1	12.9	9.7	10.0	9.6	10.3	8.4	7.3	8.3			
	taub	0.194	0.261	0.294	0.327	0.340	0.342	0.355	0.338	0.313	0.283	0.350	N/A			
	taud	1.926	2.099	2.160	2.226	2.443	2.475	2.454	2.483	2.535	2.339	2.687	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	61	181	239	262	273	275	266	260	241	179	67	0			
	Edh at Noon	5	17	27	33	30	30	30	26	19	13	5	0			
(44) All-Sky Solar Radiation	RadAvg	22	181	650	1212	1514	1625	1505	1097	572	192	40	0			
	RadStd	1	32	64	110	130	119	149	104	55	37	5	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		+1.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page